

Laparoscopic Nissen procedure in children.

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Fundoplication is among the most frequently performed procedures in infants and children. The Nissen fundoplication is the most commonly performed anti-reflux procedure done in the pediatric population. The broad indications include respiratory compromise, failure to thrive, neurologic impairment, and severe esophagitis. Over the last decade the application of a minimally invasive surgical technique to this procedure has received widespread acceptance among adult and pediatric surgeons and should now be considered the technique of choice.

Laparoscopic Nissen fundoplication.

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Open Nissen fundoplication has been established as an effective treatment in the management of gastroesophageal reflux disease (GORD). Progress in minimally invasive surgery has given the chance of performing laparoscopic Nissen fundoplication. The short-term results are encouraging and its outcome promising. The need for long-term medication ceases and it has already widely replaced the open procedure. Furthermore, the laparoscopic procedure can achieve the repair of any hiatal defect present. Thorough preoperative evaluation is imperative, because it determines the indication for surgery as well as the sort of operation. The laparoscopic procedure ensures a meticulous dissection and full mobilization of the lower oesophagus. These preconditions are of great importance in performing safely a floppy wrap. The method could be followed by complications or failure. Appropriately designed modern equipment is necessary. Thus, much experience and adequate training are required for the perfect application of laparoscopic Nissen fundoplication.

Long-term outcome of laparoscopic Nissen fundoplication in a regional hospital setting.

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Laparoscopic Nissen fundoplication is considered to be the standard approach for the surgical treatment of gastroesophageal reflux disease. Various trials have assessed the outcome of the operation performed in high-volume centers, but the existing evidence regarding peripheral, low-volume hospitals is scarce. The purpose of this study is to investigate the late outcome of laparoscopic Nissen fundoplication with regard to symptom control and postoperative quality of life in a community-hospital setting. 376 patients underwent laparoscopic Nissen fundoplication in our hospital during the period of 1997-2012. Patients were asked to subjectively assess the pre- and postoperative severity of their symptoms and quality of life. Follow-up was conducted by means of a mailed questionnaire. Respondents had a median follow-up of 8.8 years (range: 1.4-17). 60.8% (101/166) patients reported complete control of heartburn. Ten patients (6%) had undergone revisional surgery. 31 respondents (18.6%) reported a new-onset dysphagia postoperatively. 85% (138/166) of the respondents would have the operation again, if necessary. 73.6% (120/166) reported a lasting improvement of their overall quality of life. Laparoscopic Nissen fundoplication is a safe method with significant long-term efficacy in terms of symptom control and quality of life, even when performed in a low-volume, community-hospital setting.

LINX(™) Reflux Management System: magnetic sphincter augmentation in the treatment of gastroesophageal reflux disease.

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Gastroesophageal reflux disease (GERD), commonly manifested by heartburn or regurgitation, is a chronic, progressive condition in which failed sphincter function allows the contents of the stomach to reflux into the esophagus, the airways and the mouth. Chronic GERD affects 10% of Western society. The majority of patients receive adequate relief from proton pump inhibitors, but up to 40% have incomplete relief of symptoms that cannot be addressed by increasing the dose of medications. The laparoscopic Nissen fundoplication is the surgical gold standard; however, the level of technical difficulty and its side effects have limited its use to less than 1% of the GERD population. These factors have contributed to the propensity of patients to persist with medical therapy, even when inadequate to control symptoms and complications of the disease. Consequently, a significant gap in the treatment continuum for GERD remains evident in current clinical practice. The LINX(™) Reflux Management System (Torax Medical) is designed to provide a permanent solution to GERD by augmenting the physiologic function of the sphincter barrier with a simple and reproducible laparoscopic procedure that does not alter gastric anatomy and can be easily reversed if necessary.

A comparative trial of laparoscopic magnetic sphincter augmentation and Nissen fundoplication.

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Laparoscopic magnetic sphincter augmentation (MSA) with the LINX device is a promising new therapy for the treatment of gastroesophageal reflux disease (GERD). Initial studies have demonstrated MSA to be safe and effective. However, no direct comparison between MSA and laparoscopic Nissen fundoplication (LNF), the gold standard surgical therapy for GERD, has been performed. A single institution, case-control study was conducted of MSA performed from 2012 to 2013 and a cohort of LNF matched for age, gender, and hiatal hernia size. MSA and LNF were both effective treatments for reflux with 75 and 83 % of patients, respectively, reporting resolution of GERD at short-term follow-up. Dysphagia was common following both MSA and LNF, but severe dysphagia requiring endoscopic dilation was more frequent after MSA (50 vs. 0 %, $p = 0.01$). Need for dilation did not correlate with size of the LINX device or any other examined patient factors. A trend toward decreased adverse GI symptoms of bloating, flatulence, and diarrhea was seen after MSA compared to LNF (0 vs. 33 %). MSA had a shorter operative time (64 vs. 90 min, $p < 0.01$) but other peri-operative outcomes, including pain, morbidity, and re-admissions were equivalent to LNF. MSA patients were more likely to be self-referred (58 vs. 0 %, $p < 0.001$). MSA and LNF are both effective and safe treatments for GERD; however, severe dysphagia requiring endoscopic intervention is more common with MSA. Other adverse GI side effects may be less frequent after MSA. Consideration should be paid to these distinct post-operative symptom profiles when selecting a surgical therapy for reflux disease.

A Japanese patient with the Costello syndrome.

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The Costello syndrome is characterized by dwarfism, unique cutaneous lesions, distinct facial gestalt, and mental retardation. We present a Japanese patient with the Costello syndrome. She showed high serum IgM level during the early infantile period. Nissen's fundoplication was carried out to treat severe gastroesophageal reflux. Endocrinological investigations revealed a partial deficiency of growth hormone.

Mesenteroaxial gastric volvulus in an elderly patient 5 years after Nissen fundoplication and hiatal hernia repair.

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Gastric volvulus is a rare and potentially life-threatening condition that usually presents acutely and requires immediate intervention via either endoscopic or surgical detorsion. Most often, it presents secondary to a hiatal hernia, with herniation and torsion of the stomach through the hiatus. Only a small subset of patients present with gastric volvulus after Nissen fundoplication for hiatal hernia repair. We report the case of an elderly patient with a previous hiatal hernia repair with Nissen fundoplication who presented with an intra-abdominal gastric volvulus that developed over the course of several months. The combination of her surgical history and volvulus etiology made her disease rare. She was treated with laparoscopic surgery and gastropexy with gastrostomy tube placement, which led to the resolution of the upper gastrointestinal symptoms.

A Rare Case of Internal Gastroduodenal Hernia Through the Fundoplication Wrap Two Years Following Laparoscopic Nissen Fundoplication for the Treatment of Gastroesophageal Reflux Disease.

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BACKGROUND Gastroesophageal reflux disease (GERD) is a common condition that may be refractory to medical treatment with proton pump inhibitors (PPIs). Laparoscopic Nissen fundoplication is the recommended surgical treatment for GERD and is safe and effective. This report is of a rare case of internal gastroduodenal hernia as a late complication of laparoscopic Nissen fundoplication for the management of GERD in a 19-year-old woman. **CASE REPORT** A 19-year-old woman was admitted to the emergency department with a three-day history of epigastric pain, anorexia, and altered bowel habit. She had a history of GERD that was treated two years previously by laparoscopic Nissen fundoplication. On the most recent hospital admission, abdominal computed tomography (CT) showed an internal hernia of the gastroduodenal junction through the tissues used as a fundoplication wrap of the abdominal esophagus. The imaging findings were confirmed at exploratory laparoscopy, at which time surgical takedown of the fundoplication was performed. **CONCLUSIONS** This report is of a rare case of gastroduodenal hernia through a fundoplication wrap two years after a Nissen fundoplication. However, clinicians should be aware of this rare diagnosis in patients with a history of Nissen fundoplication who present with acute upper gastrointestinal symptoms.

Nissen fundoplication for gastro-oesophageal reflux disease: long-term results.

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Since its introduction by Rudolph Nissen in 1956, fundoplication has become the most commonly used antireflux procedure. Following fundoplication the majority (80 to 90%) of the patients become symptomfree or have only mild and occasional reflux symptoms in the long run. With a short and loose fundic wrap postoperative dysphagia is not a clinical problem, providing that preoperative manometry shows adequate peristalsis. Flatus is increased after fundoplication, but rarely to a disturbing extent. Patients who have problems with flatus preoperatively are also prone to have complaints postoperatively. Bloating is decreased rather than increased after fundoplication. At endoscopy as intact seen fundic wrap (the main determinant of the long-term outcome) is observed in 70% to 80% of the cases 10 to 20 years after the operation. Oesophageal 24-hour pH-recording is normal and oesophagitis cured in similar number of patients. In conclusion, Nissen fundoplication gives effective cure of symptoms of gastro-oesophageal reflux disease and reliably corrects reflux oesophagitis. Postoperative adverse effects are rare and well tolerable and do not detract from the success of the operation in correctly selected patients.

Laparoscopic Toupet fundoplication: a safe and effective anti-reflux option in lung transplant recipients.

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Fundoplication is known to improve allograft outcomes in lung transplant recipients by reducing retrograde aspiration secondary to gastroesophageal reflux disease, a modifiable risk factor for chronic allograft dysfunction. Laparoscopic Nissen fundoplication has historically been the anti-reflux procedure of choice, but the procedure is associated with discernable rates of postoperative dysphagia and gas-bloat syndrome. Laparoscopic Toupet fundoplication, an alternate anti-reflux surgery with lower rates of foregut complications in the general population, is the procedure of choice on our institution's lung transplant protocol. In this work, we evaluated the efficacy and safety of laparoscopic Toupet fundoplication in our lung transplant recipients. A prospective case series of 44 lung transplant recipients who underwent laparoscopic Toupet fundoplication by a single surgeon between September 2018 and November 2020 was performed. Preoperative and postoperative results from 24-h pH, esophageal manometry, gastric emptying, and pulmonary function studies were collected alongside severity of gastroesophageal reflux disease and other gastrointestinal symptoms. Median DeMeester score decreased from 25.9 to 5.4 after fundoplication ($p < 0.0001$), while percentage of time $\text{pH} < 4$ decreased from 7 to 1.1% ($p < 0.0001$). The severity of heartburn and regurgitation were also reduced ($p < 0.0001$ and $p = 0.0029$ respectively). Overall, pulmonary function, esophageal motility, gastric emptying, severity of bloating, and dysphagia were not significantly different post-fundoplication than pre-fundoplication. Patients with decreasing rates of FEV1 pre-fundoplication saw improvement in their rate of change of FEV1 post-fundoplication ($p = 0.011$). Median follow-up was 32.2 months post-fundoplication. Laparoscopic Toupet fundoplication provides objective pathologic acid reflux control and symptomatic gastroesophageal reflux improvement in lung transplant recipients while preserving lung function and foregut motility. Thus, laparoscopic Toupet fundoplication is a safe and effective antireflux surgery alternative in lung transplant recipients.